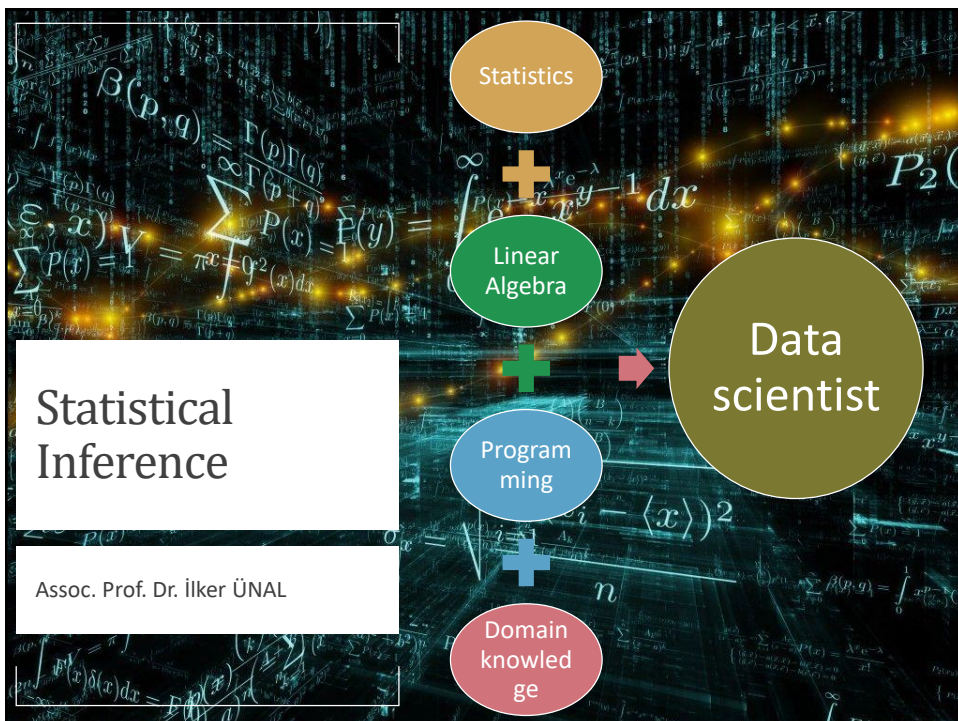
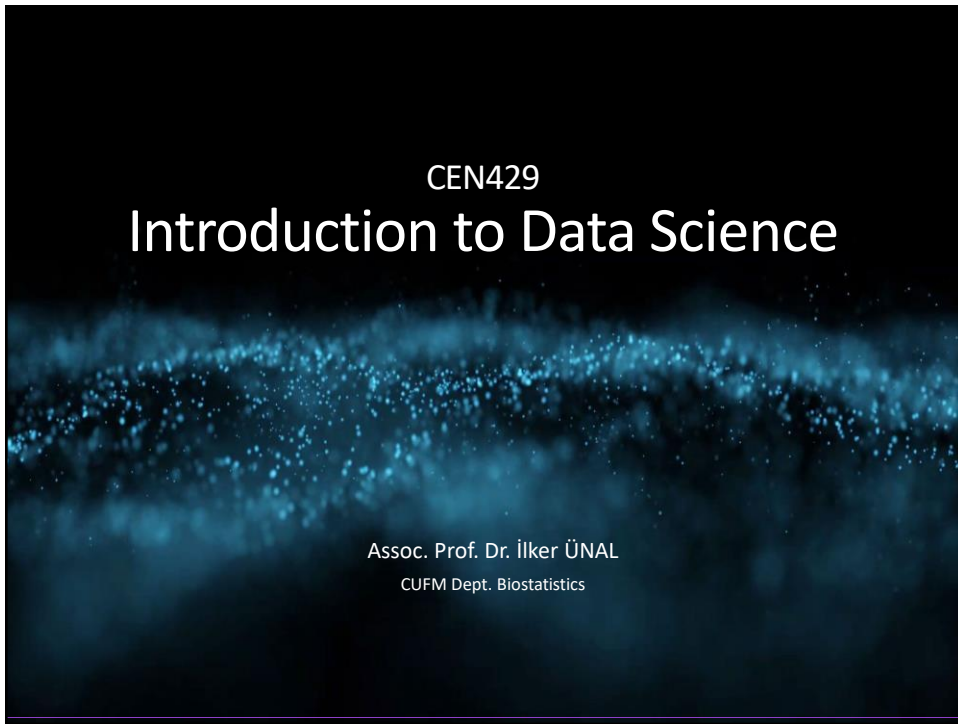


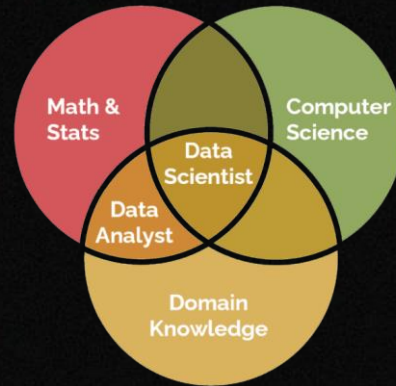


Data analyst vs Data scientist

How many of these skills do you have?

The common part is the statistics.

Why do we need to know statistics?
«statistical inference»

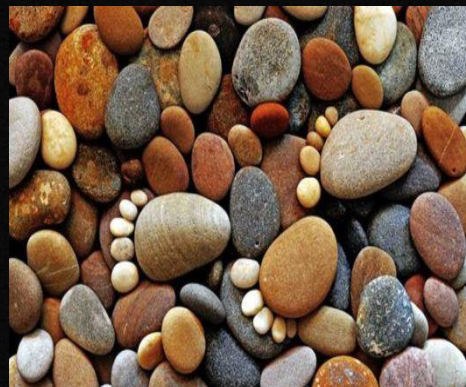


Understanding what data tells us

Why do we need to collect data?

Why using one individual data is not enough?

Do you think the information based on one individual data is **reliable**?



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Understanding what data tells us

There are two important terms
«randomness» and «uncertainty»

What is randomness?

If our knowledge about the outcome is limited



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What is uncertainty?

Not knowing the exact measure



Understanding what data tells us

The average weight of all people?

Sum the weight of all people, Divide
by the number of people

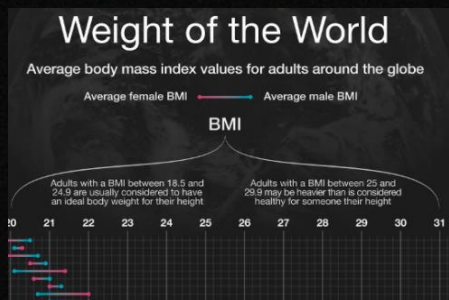
Easy to do that?

Select 10 people and calculate the
average weight

Result is exactly same?

Uncertainty can be measured

Standard error of mean (SEM)



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Data can be transformed it into knowledge

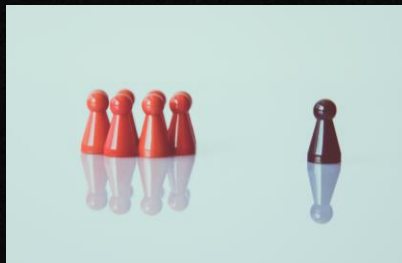


from the data to knowledge, is the field of **statistical inference**

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Populations and Samples



- Distinction between the population and the sample.
 - ✓ Turkish Population, World population
 - ✓ Only people?
 - ✓ Tweets, photographs, even all human blood in the body.
 - ✓ The capital N represents the total number

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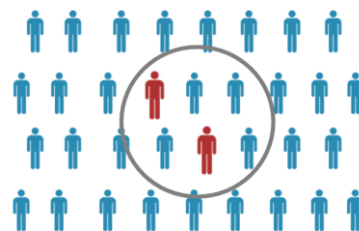


Sampling

How to get a sample
sampling mechanisms

Why the mechanism important?

- ✓ it can provide «the representation»
- ❖ it may introduce biases into the data,
- ❖ these biases may cause that any conclusions we draw will be wrong or distorted.



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Example

- The rate of smokers in the students of Cukurova University
 - ✓ More than 55000 students
 - ✓ Need to take a sample
- Way 1: one to tenth of all students
- Way 2: taking faculties as strata. From each strata, weighted sample
- Both are reasonable ways

Sample Mechanisms

In the previous example, the first scenario is an example of simple random sampling in which each **unit** has an equal probability of being chosen.

The second scenario is an example of stratified sampling which is a method of sampling from a population which can be partitioned into subpopulations.

The other sampling mechanisms are systematic sampling and cluster sampling.

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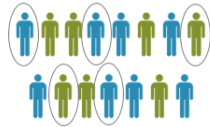
Sampling

In the random sampling, the population is being

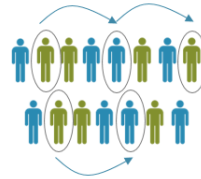
The sampling method is a simple random sampling

The sampling method is a simple random sampling

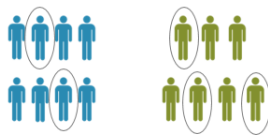
Random sampling



Systematic sampling



Stratified sampling

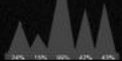


Cluster sampling



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Sampling Mechanisms

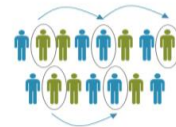
Systematic sampling assumes there is some arranging in the study population according to some ordering scheme.

Then we select elements at regular intervals through that ordered list.

If our study population is in a dispersed settlement, we may not be able to apply these mechanisms. Instead, we can take groups ('clusters') as a sample

For example, let's consider villages as clusters then taking all habitants from one village as a sample is an example of cluster sampling.

Systematic sampling



Cluster sampling



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Populations and Samples of Big Data

- In the age of Big Data, we can record all users' actions all the time

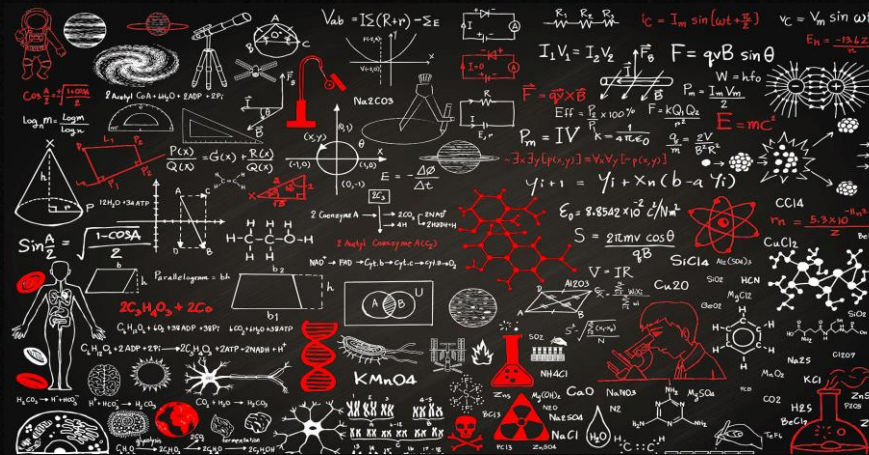


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Populations and Samples of Big Data

- In the age of Big Data, we can record all users' actions all the time



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Populations and Samples of Big Data

- How much data do we need at hand?



Populations and Samples of Big Data

- Doesn't sampling have any side effect?
 - ✓ «selection bias»



Populations and Samples of Big Data

➤ Doesn't sampling have any side effect?

✓ «selection bias»

10 5 8 12 20 11 5 9 6 14

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Populations and Samples of Big Data

➤ Doesn't sampling have any side effect?

✓ «selection bias»

5 8 12 11 9 6

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20

Populations and Samples of Big Data

- The relationship between a population and a sample
- The underlying truth, the Mathematical structure, and shape of the underlying generative process
- Only one particular realization of that generative process
 - ✓ sample
- The sampling distribution

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New kinds of data



Binary



Categorical



Numerical

Traditional



Articles



E-mail



Tweets

Text



Time
stamped



Formatted
Log

Records



Geo-based
location



Network



Sensor



Images

Other

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Modeling –

Meaning statistical models

- The uncertainty and randomness of data-generating processes express the shape and structure of the data.
- A model is necessary for understanding and representing the nature of reality.
- Constructing a model should be done carefully.
 - ✓ If something is overlooked, it may make the modelling useless.

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Statistical modeling

- Draw a picture of what you think the underlying process might be with your model.
- Express these kinds of relationships in terms of math.
- Including parameters that are not known yet.



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Statistical modeling

- In mathematical expressions,
 - ✓ Greek letters (α , β , μ ,...) for parameters and Latin letters (y , x ,...) for data.
- $y = \beta_0 + \beta_1 x$
 - ✓ β_0 and β_1 are the parameters.

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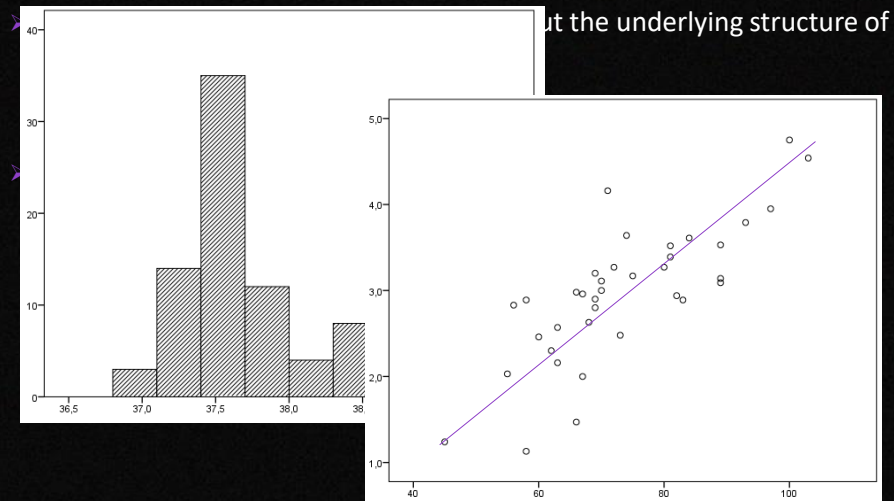
How do you build a model?

- You have to make a lot of assumptions about the underlying structure of reality
 - ✓ How?
- You can plot histograms and scatterplots to start getting a feel for the data

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How do you build a model?



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Probability Distributions

- The distribution of data generally follows a known distribution.
- Probability distributions are the foundation of statistical models.
- The famous one is normal distribution, a bell-shaped curve, also called a Gaussian distribution
- It has a probability density function



$$N(x|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

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Probability

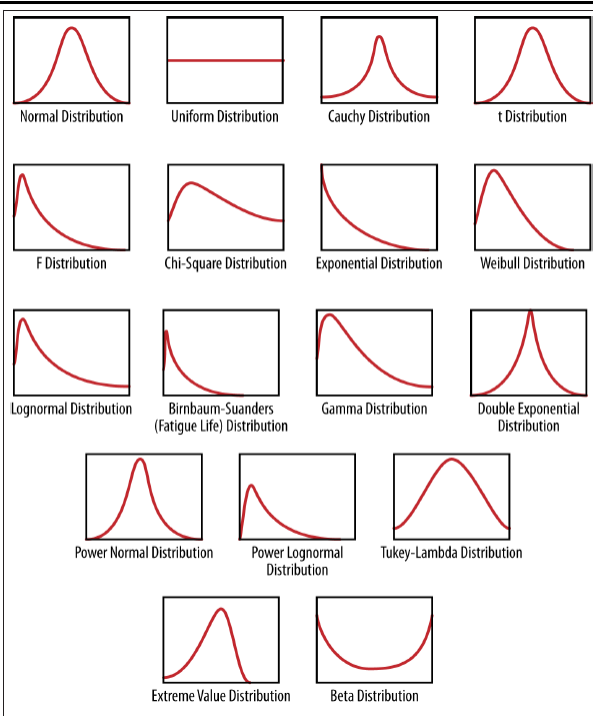
➤ The distribution

➤ Probability

➤ The famous

a bell-shaped
distribution

➤ It has a pro



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91%

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Probability Distribution – meaning

➤ A random variable denoted by x can be assumed to have a corresponding probability distribution, $p(x)$, which maps x to a positive real number.

➤ In order to be a probability density function, we are restricted to the set of functions such that if we integrate $p(x)$ to get the area under the curve, it is 1, so it can be interpreted as probability.

$$\int_{-\infty}^{\infty} p(x) dx = 1$$

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Probability Distribution – meaning

- In other words, we can calculate the probability of event using this probability density function.
- Let x be the amount of time until the next bus arrives.
 - ✓ x is a random variable, why?
- Suppose we know that the time until the next bus has a probability density function of $p(x) = 2e^{-2x}$.
- If we want to know the likelihood of the next bus arriving in between 12 and 13 minutes, then we should find the area under the curve between 12 and 13 by

$$\int_{12}^{13} 2e^{-2x} dx$$

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Probability Distribution – meaning

- How do we know this is the right distribution to use?
- There are two possible ways:
 - ✓ Conduct an experiment to measure how much time elapsed between two successive buses and repeat this measurement over and over again. Then plot measurements, and approximate the function.
 - ✓ “waiting time” measurements generally has a distribution called the exponential distribution, we know that it takes the form

$$p(x) = \lambda e^{-\lambda x}$$

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Probability Distribution – meaning

- We can use multivariate functions called *joint distributions* which are the distributions of more than one random variable.
- For example in the case of two random variables, the distribution can be

$$p(x,y)$$
- and it would take values in the plane and give us nonnegative values.
 - ✓ its (double) integral over the whole plane would be 1.
- We also have a *conditional distribution*,

$$p(x|y)$$

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Model fitting

- *Fitting* a model means «estimation of the parameters»
- Why do we estimate parameters?
 - ✓ We are using your data not all population data.
- Fitting the model = optimization methods and algorithms
 - ✓ *maximum likelihood estimation*
- To estimate the parameters, we use *estimators*, meaning *functions* of the data.
- If we fit the model, then we can write, for example, $y = 7.2 + 4.5x$

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Model fitting

- Fitting the model is when you start actually coding
 - ✓ your code will read in the data,
 - ✓ you'll specify the functional form
 - ❖ you wrote it down on the piece of paper.

- Then R or Python will use built-in optimization methods to give you the most likely values of the parameters given the data.

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R vocabulary

- **package:** Collections of R functions, data, and compiled code in a well-defined format
- **library:** The directory where packages are stored
- **object:** A specific/concrete realization of a class
 - ✓ Data objects
 - ✓ Functions
- **class:** A blueprint/template that describes the behavior/state that the object of its type can get
- **environment:** A place to store objects
- **workspace:** R working environment and includes all user-defined objects

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R properties

- Data objects
 - ✓ Vectors
 - ✓ Arrays
 - ✓ Matrices
 - ✓ Lists
 - ✓ Data frames (variables in columns, observations in rows)
- Functions
 - ✓ *functionName(arguments)*
 - ✓ `mean(x, trim = 0, na.rm = FALSE)`
- R is case sensitive!
 - ✓ Use nouns for variables, verbs for functions

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R packages

- Install the package into your computer
 - ✓ `install.packages(gtools)`
- Update all packages/one package
 - ✓ `update.packages()`
 - ✓ `update.packages(gtools)`
- Load the packages,
 - ✓ `library(gtools)`

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R functions and operators

- ? : help
 - ✓ ?mean
- $4+3$, $4/3$, 4^3 , $4\%\%3$, $4\%/%3$
- &, |, !
- <- means «equal to»
 - ✓ object_name <- value
 - ✓ a <- 5
- c(...) : This function combines terms together

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Data in R

- In R all objects have a type of information, called as «mode»
 - ✓ There are six modes
 - ❖ Logical
 - ❖ Numeric
 - ❖ Complex
 - ❖ Character
 - ❖ List
 - ❖ Function
- mode(a) # returns type of data
- b<-«a» # character
- d<-TRUE # logical
- mode(sum) # function

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Data in R

- The data can be stored in R as a vector.
- In particular there is a first element, a second element up to a last element.
 - ✓ It has a natural order.
 - ✓ You can make changes to the data item by item instead of having to enter in the entire data set again.
 - ✓ Vectors are also a mathematical object.

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R objects

- The most useful R command for quickly entering in small data sets is the `c` function.
- `a<-c(2,3,0,3,1,0,0,1)`
 - ✓ You can use also `«=»` instead of using `<-`
- To apply a function, just write the name of the function
 - ✓ `sum(a)`
- Results of functions can be stored in objects
 - ✓ `b<-sum(a)`
- Look at the structure of the object
 - ✓ `str(a)`

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Data in R

- `a<-c(2,3,0,3,1,0,0,1)`
- `a2<-a` # make a copy
- `a2[1]<-0` # assign the first entry 0
- the comment character, #, is used to make comments.
 - ✓ Anything after the comment character is ignored
- The assignment to the first entry in the vector `a2` is done with square brackets `[1]`.
 - ✓ parentheses `()` are for functions,
 - ✓ square brackets `[]` are for vectors

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Data in R

- `a2` # print out the value
- `a2[2]` # print 2nd entry value
- `a2[-4]` # all but the 4th entry
- `a2[c(1,2,3)]` # print 1st, 2nd and 3rd.
- Notice negative indices give everything except these indices.
- You can take more than one value at a time by using another vector of index numbers.
 - ✓ This is called slicing.

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Data in R

- Let's find the maximum value in this vector.
 - ✓ `max(a)`
- Which entries are equal to the maximum value?
 - ✓ `a == 3`
 - ✓ the usage of double equals signs (`==`)
- How can we get the indices corresponding to the TRUE values?
 - ✓ `which(a==3)`
- To find the number of the entries in the vector
 - ✓ `length(a)`

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Data in R

- To create the vector 1 2 3 ... we can use the simple `:` colon operator.
 - ✓ `a3<-c(1:8)`
- Now, what if you didn't think of the command `which`?
 - ✓ `n<-length(a2)`
 - ✓ `ids<-1:n`
 - ✓ `ids[a2==3]`
- The command `a:b` is simply `a`, `a+1`, `a+2`, ..., `b` if `a`, `b` are integers
 - ✓ More general form `seq(a,b,1)`
 - ✓ `seq(10,20,2)`

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Data in R

- Knowing how to use slicing and logical vectors gives you the ability to easily access your data as you desire
 - ✓ `(1:length(a2))[a2 == max(a2)]`
- If we want to know the number of entries whose values are greater than 0, we first identify the entries with values greater than 0.
 - ✓ `a2>0`
 - ✓ `sum(a2>0)`

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Data in R

- Append values to a vector
 - ✓ `x<-c(45,43,46,48,51,46,50,47,46,45)`
 - ✓ `length(x)`
 - ✓ `x<-c(x,48,49,51,50,49)`
 - ✓ `length(x)`
- Another ways to append
 - ✓ `x[16]<-41`
 - ✓ `x[17:20]<-c(40,38,35,40)`

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Data in R

➤ Taking sample

- ✓ `sample(c(1:10),size=10,replace=TRUE)`
- ✓ `sample(c(1:10), size = 5, replace = FALSE)`

➤ To create other types of vectors

- ✓ `a<-c(TRUE, FALSE, T, F)` # logical (true/false) vector
- ✓ `b<-c(«Ceng»,«cen429»)` # vector of characters
- ✓ `class(b)` # type of vector (`mode(b)`, `typeof(b)`,`str(b)`)

➤ The other type of data objects

- ✓ `d<-matrix(c(1:12),nrow=4,ncol=3,byrow=TRUE)`

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Data in R

➤ The other type of data objects

- ✓ `a <- list(a=c(1:10), b = c("a"), c = c(TRUE, FALSE, TRUE, FALSE))`
- ✓ `summary(a)`
- ✓ `aa <- array(c(1:12), dim = c(6, 2))` # same as matrix with 6 rows and 2 columns
- ✓ `bb <- array(c(1:24), dim = c(4, 3, 2))` # multidimensional object
- ✓ `e <- data.frame(a = c(1:5), b = rnorm(5), d = c("a", "b", "c", "d", "e"))`
- ✓ `names(e)`
- ✓ `e<-cbind(e,F=c(6:10))`
- ✓ `e<-rbind(e,c(6,0.5,«f»,11))`
- ✓ `e$F<-NULL`

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Data in R

- There are some other ways to edit data that use a spreadsheet interface.
 - ✓ `data.entry(x)`
- You can edit a vector if the vector is defined previously.
 - ✓ If it is not defined, you can define it as NA
 - ✓ `data.entry(x=c(NA))`
- You can see all the variables saved in your workspace by typing
 - ✓ `ls()`
- To remove an object
 - ✓ `rm(a)`
 - ✓ `rm(list=ls())` # remove all objects in the environment

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To sum up

- Suppose `x` is the data vector, for example `x<-c(1:10)`.

how many elements?	<code>length(x)</code>
ith element	<code>x[2]</code> ($i = 2$)
all but ith element	<code>x[-2]</code> ($i = 2$)
first k elements	<code>x[1:5]</code> ($k = 5$)
last k elements	<code>x[(length(x)-5):length(x)]</code> ($k = 5$)
specific elements.	<code>x[c(1,3,5)]</code> (First, 3rd and 5th)
all greater than some value	<code>x[x>3]</code> (the value is 3)
bigger than or less than some values	<code>x[x< -2 x > 2]</code>
which indices are largest	<code>which(x == max(x))</code>

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Exercises

- How do you imagine the structure of this data?
- ✓ How many variables are there?

	1st grade	2nd grade	3rd grade	4th grade
Faculty of Medicine	399	457	379	338
Faculty of Engineering	641	651	684	845
Faculty of Education	920	847	966	1011
Faculty of Agriculture	288	261	271	441

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Exercises

- How do you imagine the structure of this data?
- ✓ How many variables are there?

	1st grade	2nd grade	3rd grade	4th grade
Faculty of Medicine	399	457	379	338
Faculty of Engineering	641	651	684	845
Faculty of Education	920	847	966	1011
Faculty of Agriculture	288	261	271	441
...				

There are 3 variables and $4 \times 4 = 16$ observations
Faculty, Grade and the number of students

Faculty	Grade	Students
Medicine	1	399
Medicine	2	457
Medicine	3	379
Medicine	4	338
Engineering	1	641
Engineering	2	651
...		

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Exercises

- What is the sum of the first 100 positive integers? The formula for the sum of integers 1 through n is $n(n + 1)/2$. Define $n = 100$ and then use R to compute the sum of 1 through 100 using the formula. What is the sum?
- What is the sum of two digit odd positive integers?
- For the first 100 positive integers, how many integers can be divided both 4 and 5 at same time?

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